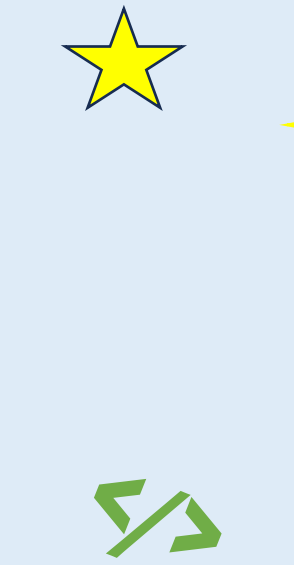
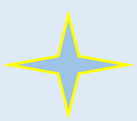
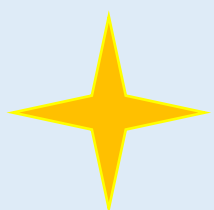
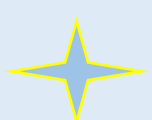


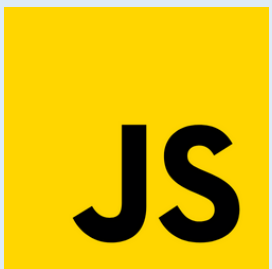






" Learn Programming For Beginners – Free Full Course "

#3 – Javascript



Programming

Programming is the way we give instructions to a computer to perform specific tasks.

The instructions must be correct, well-structured, follow rules of the programming language being used.

Why Javascript ????

- The Programming Language of the web
- Free
- Easy to Understand
- Versatile

History Javascript

1992

National Center for Supercomputing Applications (NCSA), University of Illinois at Urbana–Champaign developed Mosaic,

1994

The Mosaic team created Netscape and launched Netscape Navigator

1995

Brendan Eich get hired

Created A New language inspire by Java + Scheme + Self = Mocha (in just 10 days!)

Mocha -> Livescript -> Javascript

1996

Netscape sought Standardization

W3C rejected the proposal, so Netscape turned to ECMA International

ECMAScript

ECMAScript 1 (1997) -> First Edition

ECMAScript 2 (1998) -> Only Editorial Changes

ECMAScript 3 (1999) -> Regex, Try/Catch, Switch, do-while

ECMAScript 4 -> Never Released

ECMAScript 5 (2009) -> use strict, JSON Support, String.trim(), Array.isArray(), Array iteration methods, Allows trailing commas for object literals

ECMAScript 6 (2015) -> Added let and const, Default parameter values, Array.find(), Array.findIndex()

How To Add

```
<script src=“./script.js”></script>
```

```
<script>//your code </script>
```

it can be placed in the <body>, or in the <head> section of an HTML page

Output

```
console.log(“output”)
```

```
document.getElementById("demo").
innerText = “demo”
```

```
document.write(“output”)
```

```
window.alert(“output”)
```

Variable

Variable = containers for storing data

Const

Let

Var (obsolete)

Automatically (obsolete)

Rules Variable :

- Variable can contain letter, digits, underscores, and dollarsign
- Variable must begin with a letter, a $ sign or an underscore
- Hyphens are not allowed in Javascript
- JavaScript programmers tend to use lower camel case
  example :
  const yourName = “Josse Surya Pinem

8 DataTypes

String = A Text of Characters enclosed
example :
const person = “Josse Surya Pinem”;

Number = A number representing a mathematical value
example :
const length = 18.95;

Bigint = A number representing a large integer
example :
const x = 1234567890123456789012345n;

Boolean = A data type representing true of false
example :
const correct = true;

Object = A collection of key-value pairs of data
example :

```
const employee = {
 name: "Josse Surya Pinem",
 position: "Software Developer"
};
```

Undefined = A primitive variable with no assigned value
example :

```
let x;
```

Null = A primitive value representing object absence
example :

```
let x = null
```

Symbol = A unique and primitive indentifier
example :

```
const x = Symbol()
```

Operator

Operators are for Mathematical and Logical Computations

+ = Addition

- = Subtraction

* = Multiplication

/ = Division

% = Modulus (divison of remainder)

-- = decrement

++ = increment

== = equal to

=== = equal value dan equal data type

!= = not equal

!== = not equal value and not equal data type

> = greater than

>= = greater than or equal to

< = less than

<= = less equal than or equal to

&& = and

|| = OR

Conditionals

Conditional Statements allow us to perform different actions for different conditions

Conditional Statements run different code depending on true or false conditions

Syntax

```
If (condition1) {
  // code to execute if condition1 is true
} else if (condition2){
  // code to execute if the condition1 is false and condition2 is true
} else {
 // code to execute if the condition1 is  false, and condition2 is false
}
```

```
switch(expression){
 case x:
  //code block
  break;
 case y:
  //code block
  break;
 default:
   //code block
}
```

```
condition ? expression1 : expression2
```

Loops

Loops are handy, if you want **to run the same code over and over again**, each time with a different value

script.js

```
console.log(1);
console.log(2);
console.log(3);
console.log(4);
console.log(5);
```

script.js

```
for (let i = 1; i < 6; i++){
    console.log(i);
}
```

script.js

```
let i = 0;
do{
 console.log(i)
 i++;
}
while ( i < 6 );
```

script.js

```
let i = 0;
while (i < 6){
 console.log(i)
 i++;
}
```

Function

Functions are fundamental building blocks in all programming
Functions are resuable block of code designed to perform a particular task
Functions are executed when they are called invoked

Syntax :

```
function myFunction(param){
 // code to be executed
}
```

Syntax :

```
const myFunction = (param) =>{
 // code to be executed
}
```

```
// call or invoke function
myFunction(args)
```

Objects

An Object is a variable that can hold many variables
Objects are collections of **key-value pairs,** where each key (known as property name) has a value
Objects can describe anything like houses, cars, people, animals, or any other subjects
Objects are container for **Properties and Method**

example

```
const person = {
  firstName : “Josse”,
  lastName : “Pinem”,
  id: 1,
  fullName: function(){
   return this.firstName + “ ”
    + this.lastName
  }
}
```

Property

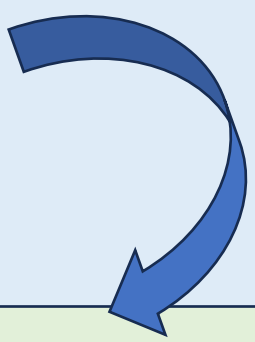
Method

Array

An Array is an object type designed for storing data collections

example:

```
const lecturer1 = “josse”
const lecturer2 = “surya”
const lecturer3 = “pinem”
const lecturer4 = “john”
```



```
const lecterures = [“josse”, “surya”,“pinem”,“john”];
```

Array Method

push() = method adds new items to the end of an array

filter() = creates a new array with every element in an array that pass a test

map() = creates a new array with the result of calling a function for each array element

splice() = add or removes array elements
array.splice(index, count, item1, ..., itemX)
Index = position array
count = many of items to be placed or removed
Item1 = element to be added

reverse() = reverses the order of the elements in an array

findIndex() = it returns that index of the first element that passes the test.

split() = it returns that An array containing the splitted values.

DOM
(Document Object Model)

The HTML DOM is a standard for how to get, change, add, or delete HTML elements.

The HTML DOM is a standard for how to find and manipulate HTML elements – Josse

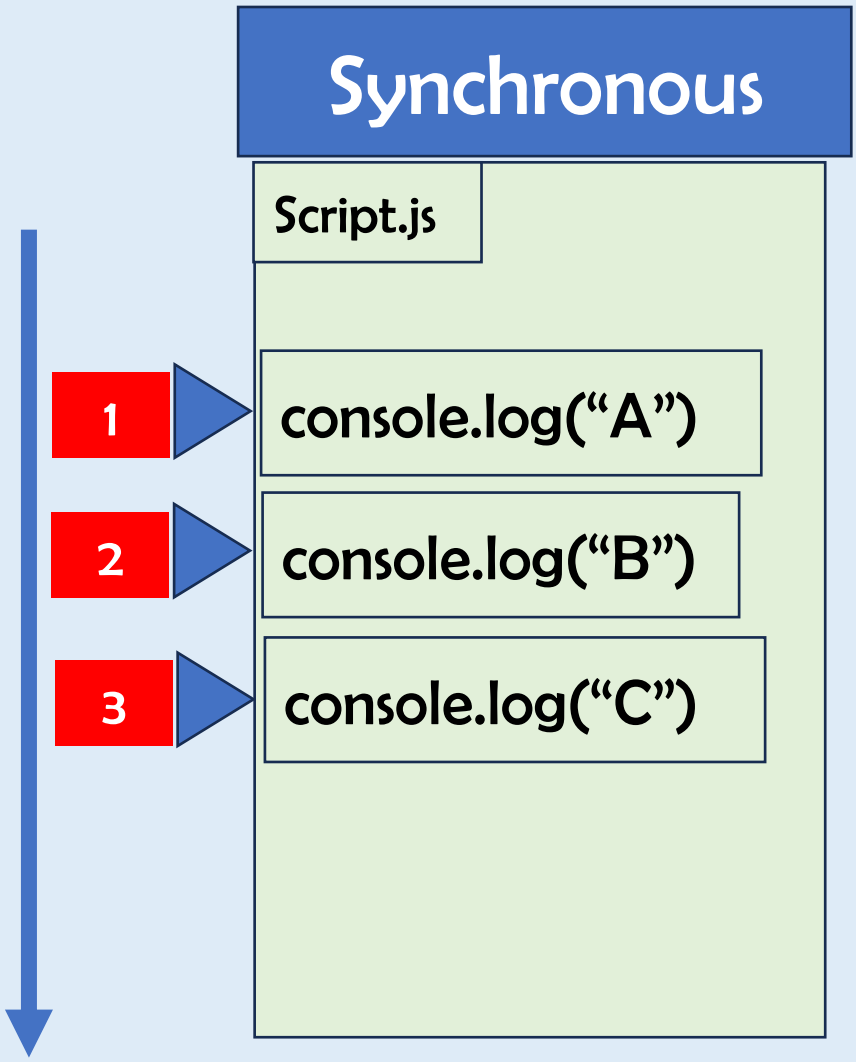
Finding HTML elements :
document.querySelector()
document.getElementById()
document.getElementsByTagName()
document.getElementsByClassName()

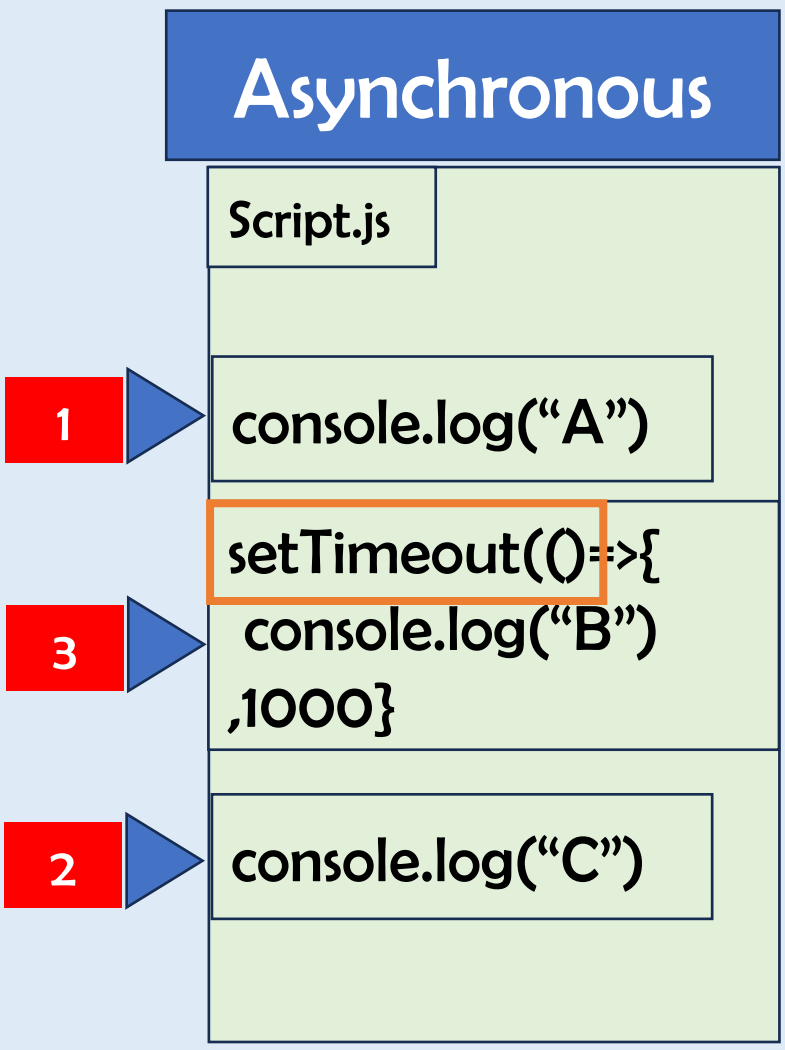
Changing elements :
element.innerHTML = *new html content*
element.style.property = *new style*
element.classList.add|remove|toggle.(“selectorclass)

Adding Events Handlers :
element.addEventListener(“click”, myFunction)

Adding Event Handlers Drag :

1. dragstart
2. dragover
3. dragenter
4. dragleave
5. drop
6. dragend


Synchronous
Script.js
1
console.log("A")
2
console.log("B")
3
console.log("C")


Asynchronous
Script.js
1
console.log("A")
setTimeout(()=>{
console.log("B")
,1000}
3
2
console.log("C")

Callback

A callback is a **function passed as an argument** to another function

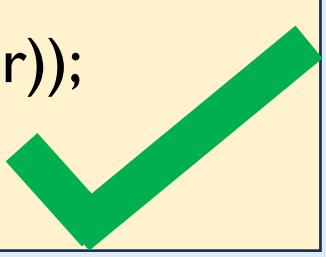
example

```
const  getData = (callback) => {
    setTimeout(()=>{
        callback("Data Loaded")
     }, 1000);
}
getData((result))=>{
  console.log(result)
});
```

```
getUser(id, (user) => {
  getPosts(user, (posts) => {
    getComments(posts, (comments)
  => {
        processComments(comments, (result)
        =>{console.log(result)});
    });
  });
});
```

Callback Hell !!!

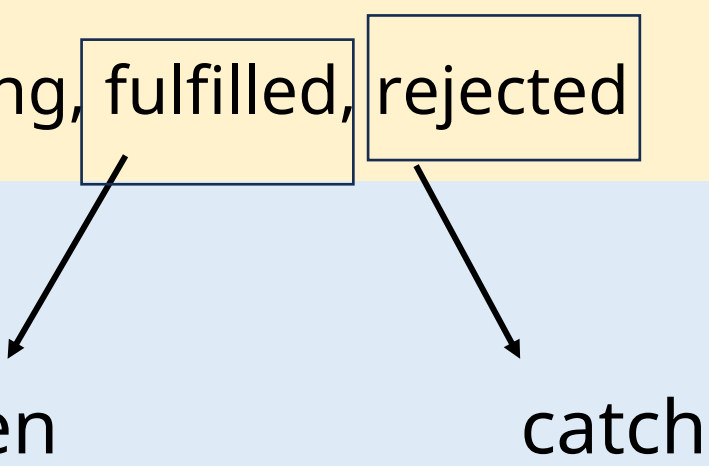
```
getUser(id)
.then(user => getPosts(user))
.then(posts => getComments(posts))
.then(comments =>{
        processComments(comments))}
.then(result => console.log(result))
.catch(error => console.error(error));
```



Promise

A object represents a value that will be available now, later, or never

States = pending, fulfilled, rejected



Example Promise & Fetch API

```
const getUser = () => {
  fetch(`https://fakestoreapi.com/users`)
  .then(res => response.json())
  .then(res => console.log(res))
  .catch(err => console.log(err))
}
getUser()
```

Promise Fetch API with async await

```
const getUser = async () => {
  try {
    const res = await
    fetch(`https://fakestoreapi.com/users`)
    const data = await response.json()
    console.log(data)
     } catch (err) {
      console.error(err)
  }
 }
getUser()
```

References :

https://www.w3schools.com/js/default.asp

THANK YOU !

Don't forget to visit

https://josserich.github.io